Inverter MCU module

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Legend

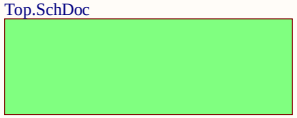
General comment such as function title, configuration, ...

Text to be added to silkscreen.

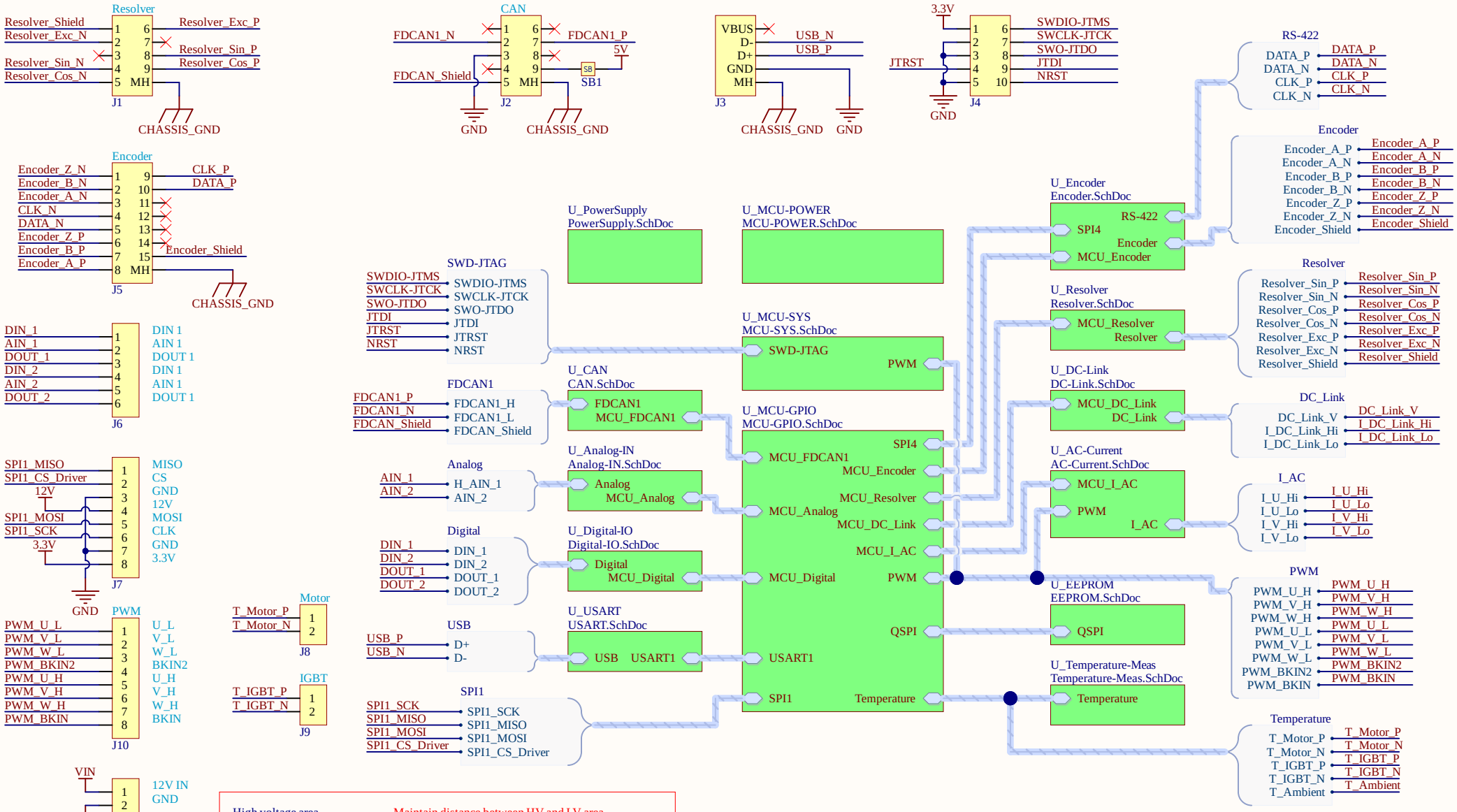
Warning text.

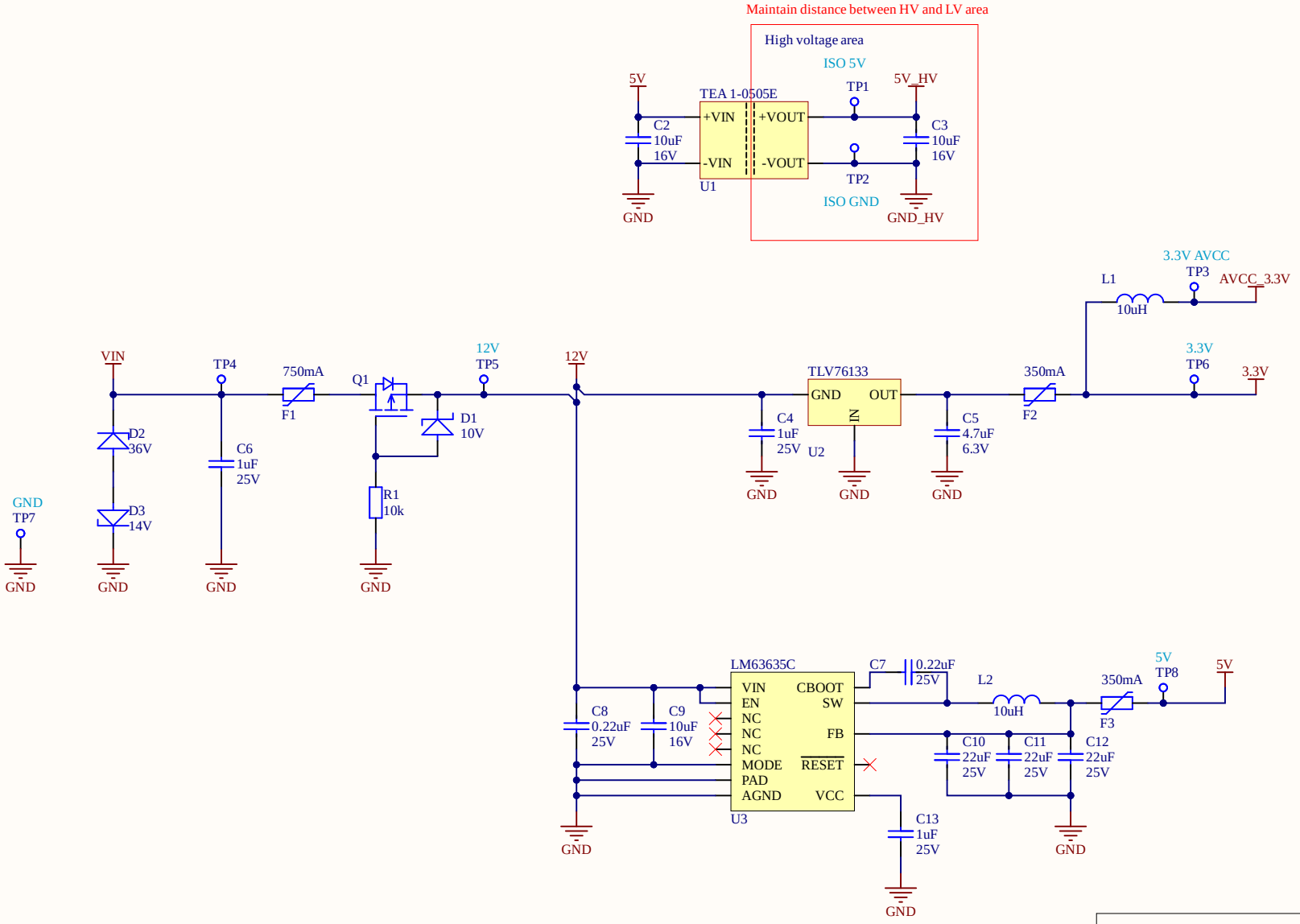
Notes to generate the board layout.

U_Top
Top.SchDoc

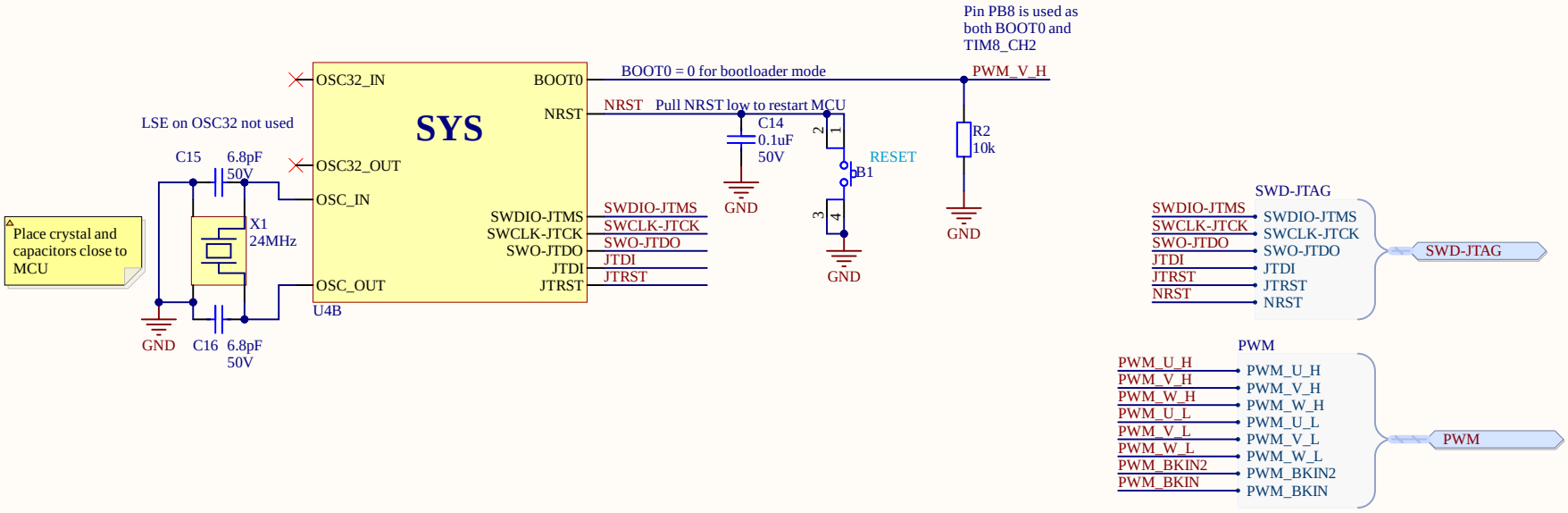


Project: MCU_module.PrjPcb		
Title: Project_overview.SchDoc		
FESB RACING	Size: A4	Revision: V1.0
Drawn by: Šimun Batovanja		Created: 16-Jul-25
Reviewed by: =CheckedBy		Edited: 07-Oct-25

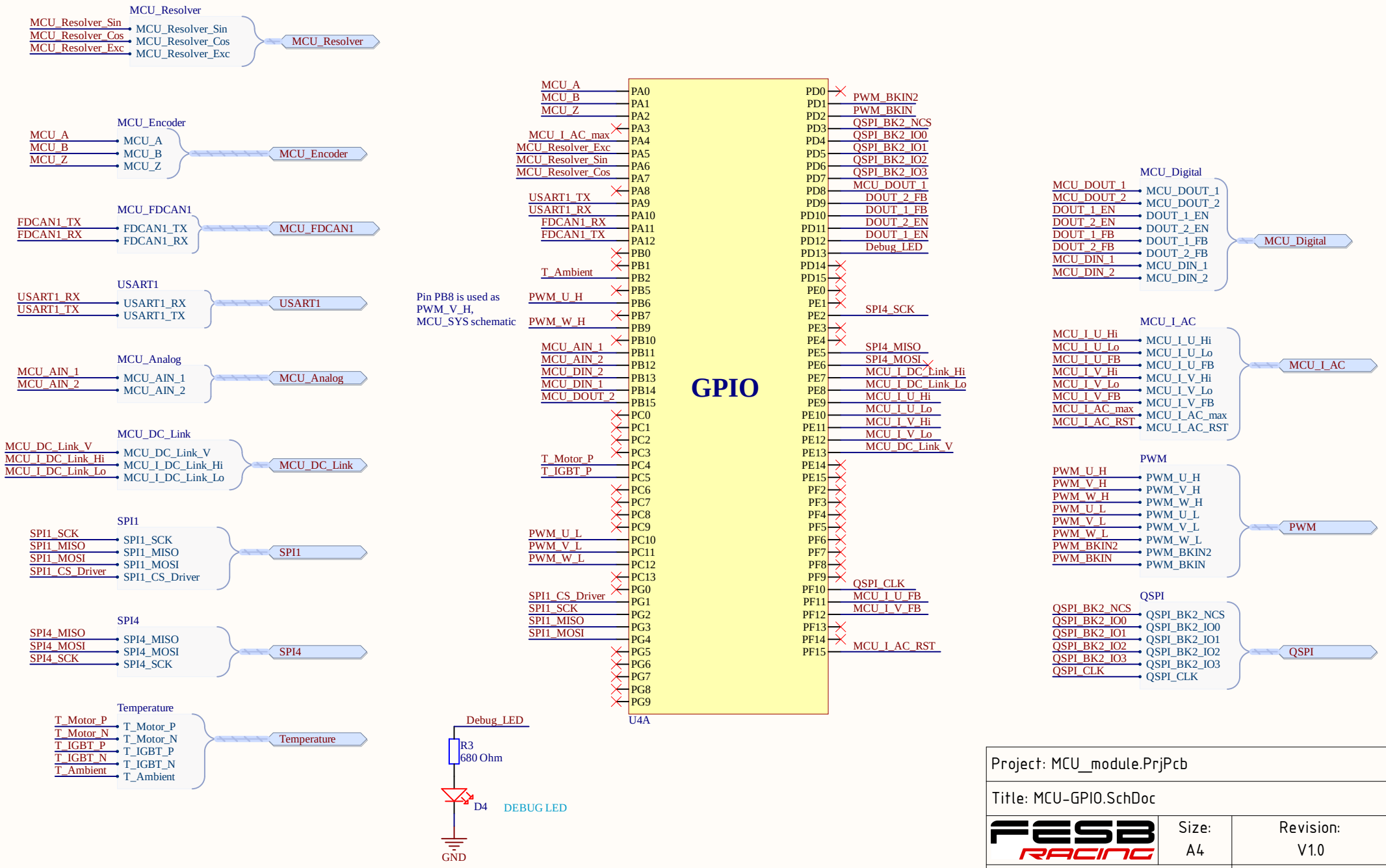

Resolver
Resolver_Shield
Resolver_Exc_N
Resolver_Sin_N
Resolver_Cos_N
Resolver_Exc_P
Resolver_Sin_P
Resolver_Cos_P
J1
CHASSIS_GND
CAN
FDCAN1_N
FDCAN_Shield
FDCAN1_P
5V
SB1
J2
GND
CHASSIS_GND
VBUS
D-
D+
GND
MH
USB_N
USB_P
J3
CHASSIS_GND
GND
3.3V
JTRST
SWDIO-JTMS
SWCLK-JTCK
SWO-JTDO
JTDI
NRST
J4
GND
Encoder
Encoder_Z_N
Encoder_B_N
Encoder_A_N
CLK_N
DATA_N
Encoder_Z_P
Encoder_B_P
Encoder_A_P
CLK_P
DATA_P
Encoder_Shield
J5
CHASSIS_GND
DIN_1
AIN_1
DOUT_1
DIN_2
AIN_2
DOUT_2
DIN 1
AIN 1
DOUT 1
DIN 1
AIN 1
DOUT 1
J6
SPI1_MISO
SPI1_CS_Driver
12V
SPI1_MOSI
SPI1_SCK
3.3V
MISO
CS
GND
12V
MOSI
CLK
GND
3.3V
J7
GND
PWM
PWM_U_L
PWM_V_L
PWM_W_L
PWM_BKIN2
PWM_U_H
PWM_V_H
PWM_W_H
PWM_BKIN
U_L
V_L
W_L
BKIN2
U_H
V_H
W_H
BKIN
J10
Motor
T_Motor_P
T_Motor_N
J8
IGBT
T_IGBT_P
T_IGBT_N
J9
VIN
12V IN
GND
J11
GND
I_V_Hi
I_U_Hi
I_DC_Link_Hi
I_V_Lo
I_U_Lo
I_DC_Link_Lo
I_V_Hi
I_U_Hi
I_DC_Hi
I_V_Lo
I_U_Lo
I_DC_Lo
J12
U_PowerSupply
PowerSupply.SchDoc
U_MCU-POWER
MCU-POWER.SchDoc
SWD-JTAG
SWDIO-JTMS
SWCLK-JTCK
SWO-JTDO
JTDI
JTRST
NRST
U_MCU-SYS
MCU-SYS.SchDoc
SWD-JTAG
PWM
FDCAN1
FDCAN1_P
FDCAN1_N
FDCAN_Shield
FDCAN1_H
FDCAN1_L
FDCAN_Shield
U_CAN
CAN.SchDoc
FDCAN1
MCU_FDCAN1
U_MCU-GPIO
MCU-GPIO.SchDoc
SPI4
MCU_FDCAN1
MCU_Encoder
MCU_Resolver
MCU_Analog
MCU_DC_Link
MCU_I_AC
MCU_Digital
PWM
QSPI
USART1
SPI1
Temperature
Analog
AIN_1
AIN_2
H_AIN_1
AIN_2
U_Analog-IN
Analog-IN.SchDoc
Analog
MCU_Analog
Digital
DIN_1
DIN_2
DOUT_1
DOUT_2
DIN_1
DIN_2
DOUT_1
DOUT_2
U_Digital-IO
Digital-IO.SchDoc
Digital
MCU_Digital
USB
USB_P
USB_N
D+
D-
U_USART
USART.SchDoc
USB
USART1
SPI1
SPI1_SCK
SPI1_MISO
SPI1_MOSI
SPI1_CS_Driver
SPI1_SCK
SPI1_MISO
SPI1_MOSI
SPI1_CS_Driver
U_Encoder
Encoder.SchDoc
RS-422
SPI4
Encoder
MCU_Encoder
U_Resolver
Resolver.SchDoc
MCU_Resolver
Resolver
U_DC-Link
DC-Link.SchDoc
MCU_DC_Link
DC_Link
U_AC-Current
AC-Current.SchDoc
MCU_I_AC
PWM
I_AC
U_EEPROM
EEPROM.SchDoc
QSPI
U_Temperature-Meas
Temperature-Meas.SchDoc
Temperature
RS-422
DATA_P
DATA_N
CLK_P
CLK_N
DATA_P
DATA_N
CLK_P
CLK_N
Encoder
Encoder_A_P
Encoder_A_N
Encoder_B_P
Encoder_B_N
Encoder_Z_P
Encoder_Z_N
Encoder_Shield
Encoder_A_P
Encoder_A_N
Encoder_B_P
Encoder_B_N
Encoder_Z_P
Encoder_Z_N
Encoder_Shield
Resolver
Resolver_Sin_P
Resolver_Sin_N
Resolver_Cos_P
Resolver_Cos_N
Resolver_Exc_P
Resolver_Exc_N
Resolver_Shield
Resolver_Sin_P
Resolver_Sin_N
Resolver_Cos_P
Resolver_Cos_N
Resolver_Exc_P
Resolver_Exc_N
Resolver_Shield
DC_Link
DC_Link_V
I_DC_Link_Hi
I_DC_Link_Lo
DC_Link_V
I_DC_Link_Hi
I_DC_Link_Lo
I_AC
I_U_Hi
I_U_Lo
I_V_Hi
I_V_Lo
I_U_Hi
I_U_Lo
I_V_Hi
I_V_Lo
PWM
PWM_U_H
PWM_V_H
PWM_W_H
PWM_U_L
PWM_V_L
PWM_W_L
PWM_BKIN2
PWM_BKIN
PWM_U_H
PWM_V_H
PWM_W_H
PWM_U_L
PWM_V_L
PWM_W_L
PWM_BKIN2
PWM_BKIN
Temperature
T_Motor_P
T_Motor_N
T_IGBT_P
T_IGBT_N
T_Ambient
T_Motor_P
T_Motor_N
T_IGBT_P
T_IGBT_N
T_Ambient
High voltage area
Maintain distance between HV and LV area
DC_Link_V
HV+
M5
Reinforced insulation
M6
HV-
GND_HV
- Clearance according to IPC-2221 Table 6-1, with coating: 0.4mm
- Creepage according to IEC 60664-1 Table F.4, Pollution Degree 3, Material Group IIIa: 3.2mm
M1 M2 M3 M4
CHASSIS_GND
Class Y capacitor
CHASSIS_GND
C1
0.01uF
1000V
GND
Inverter MCU module
Šimun Batovanja
2025 / 2026
Project: MCU_module.PrjPcb
Title: Top.SchDoc
FESB RACING
Size: A4
Revision: V1.0
Drawn by: Šimun Batovanja
Created: 16-Jul-25
Reviewed by: =CheckedBy
Edited: 07-Oct-25



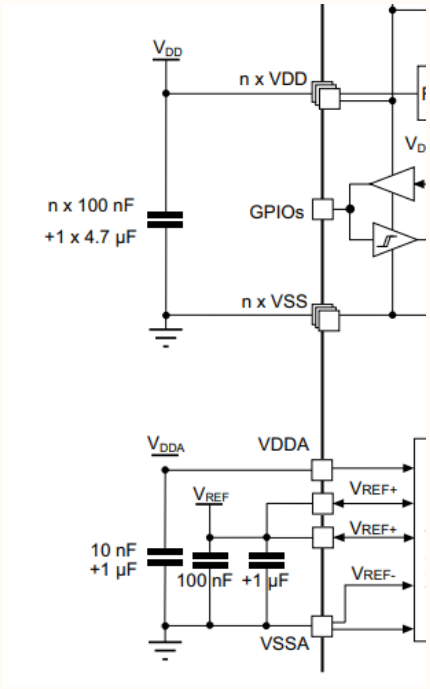
Project: MCU_module.PrjPcb		
Title: PowerSupply.SchDoc		
FESB RACING	Size: A4	Revision: V1.0
Drawn by: Šimun Batovanja		Created: 16-Jul-25
Reviewed by: =CheckedBy		Edited: 07-Oct-25

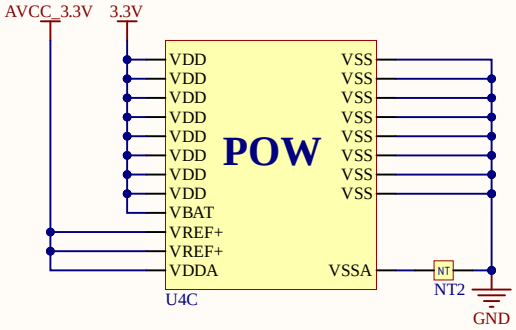


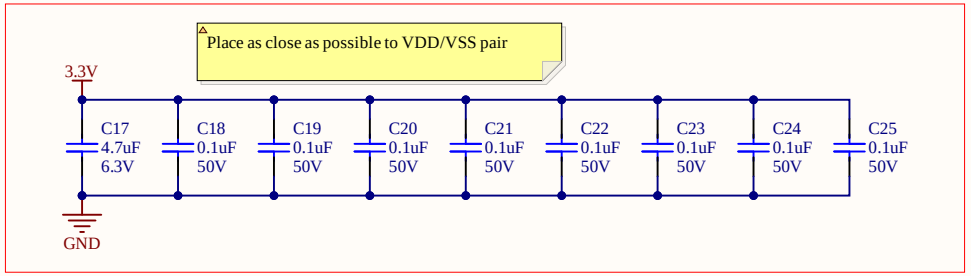
Project: MCU_module.PrjPcb		
Title: MCU-SYS.SchDoc		
FESB RACING	Size: A4	Revision: V1.0
Drawn by: Šimun Batovanja		Created: 16-Jul-25
Reviewed by: =CheckedBy		Edited: 06-Oct-25

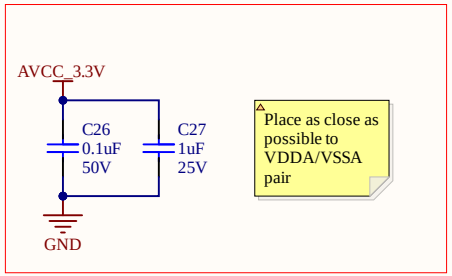


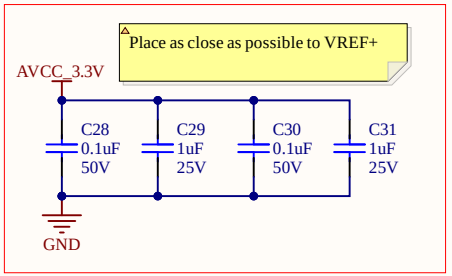
Project: MCU_module.PrjPcb		
Title: MCU-GPIO.SchDoc		
FESB RACING	Size: A4	Revision: V1.0
Drawn by: Šimun Batovanja		Created: 16-Jul-25
Reviewed by: =CheckedBy		Edited: 06-Oct-25





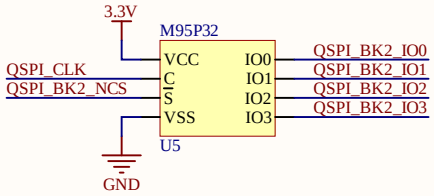


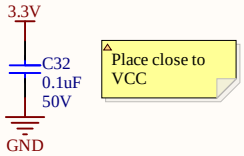




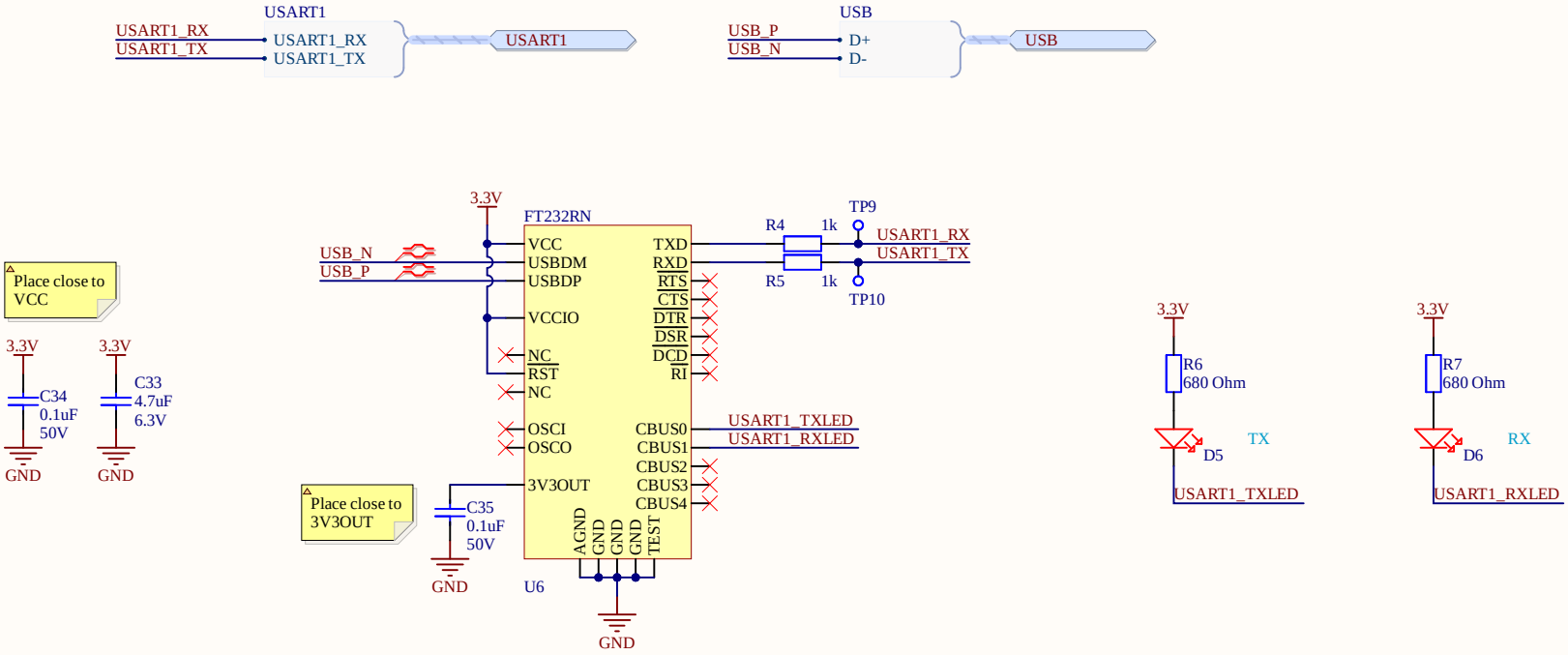
Project: MCU_module.PrjPcb		
Title: MCU-POWER.SchDoc		
FESB RACING	Size: A4	Revision: V1.0
Drawn by: Šimun Batovanja		Created: 16-Jul-25
Reviewed by: =CheckedBy		Edited: 06-Oct-25



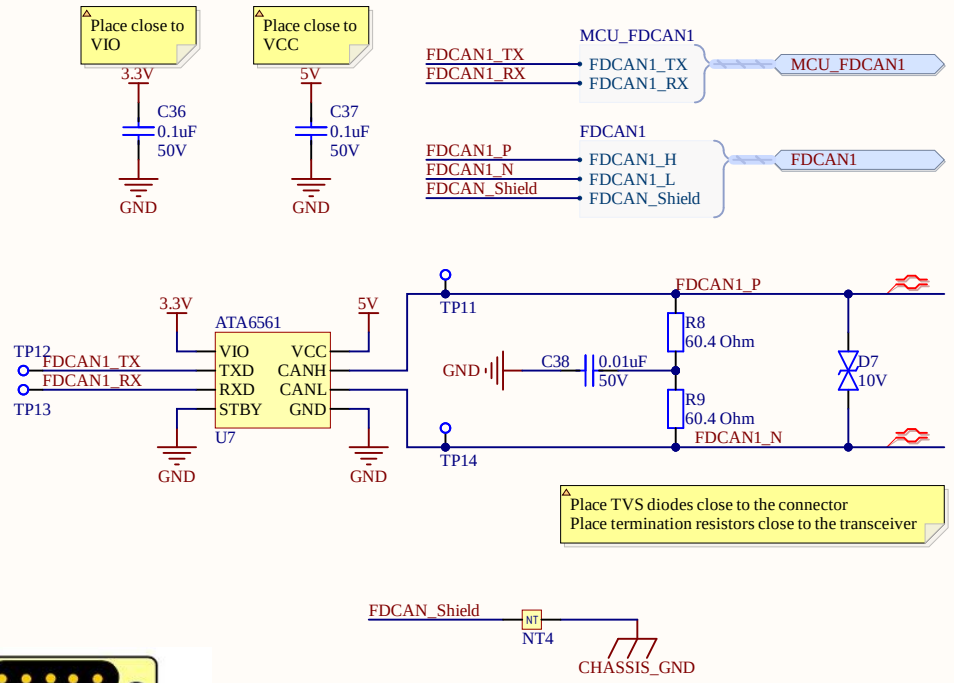




Project: MCU_module.PrjPcb		
Title: EEPROM.SchDoc		
FESB RACING	Size: A4	Revision: V1.0
Drawn by: Šimun Batovanja		Created: 16-Jul-25
Reviewed by: =CheckedBy		Edited: 06-Oct-25



Project: MCU_module.PrjPcb		
Title: USART.SchDoc		
FESB RACING	Size: A4	Revision: V1.0
Drawn by: Šimun Batovanja		Created: 16-Jul-25
Reviewed by: =CheckedBy		Edited: 06-Oct-25

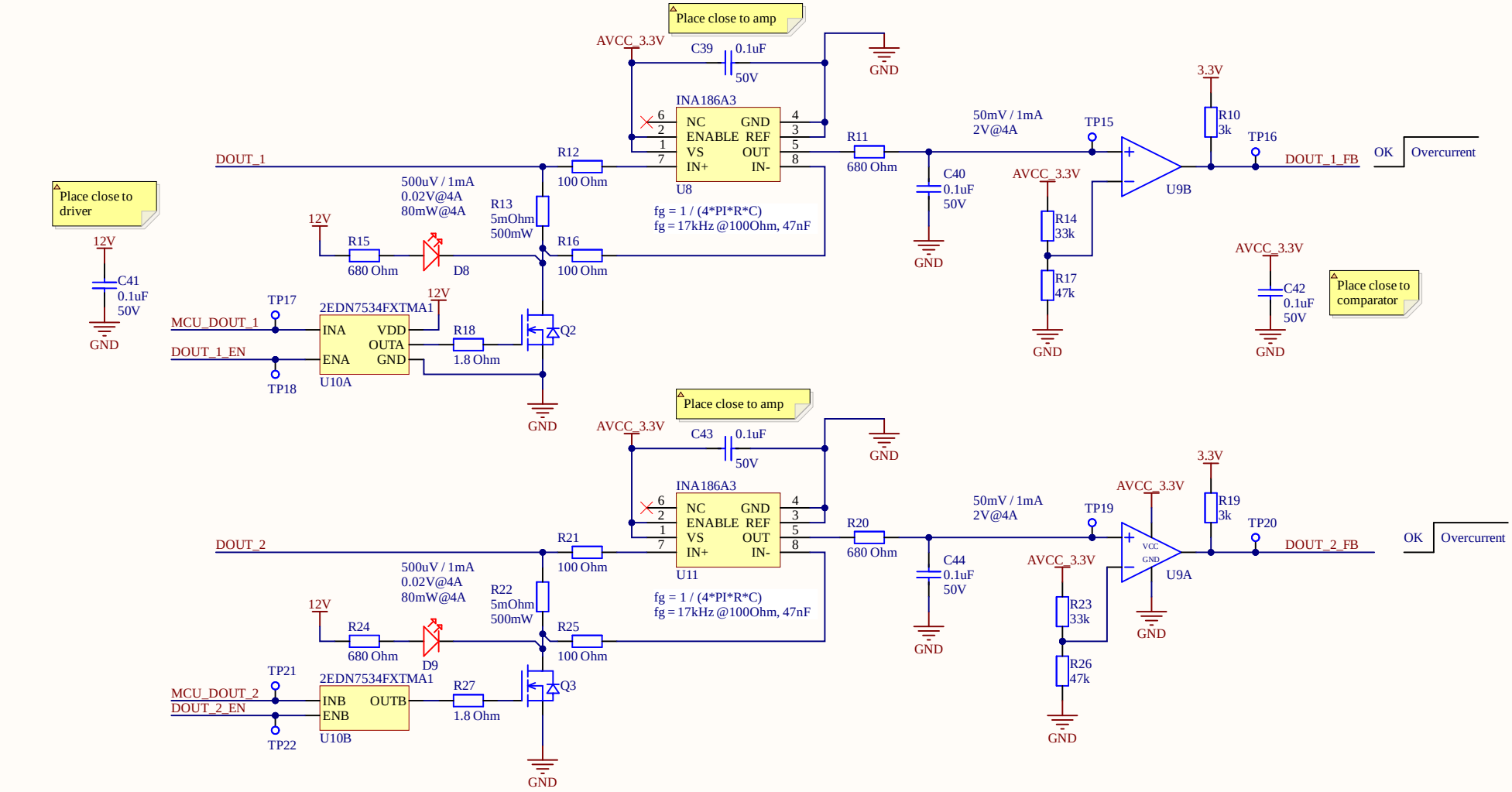




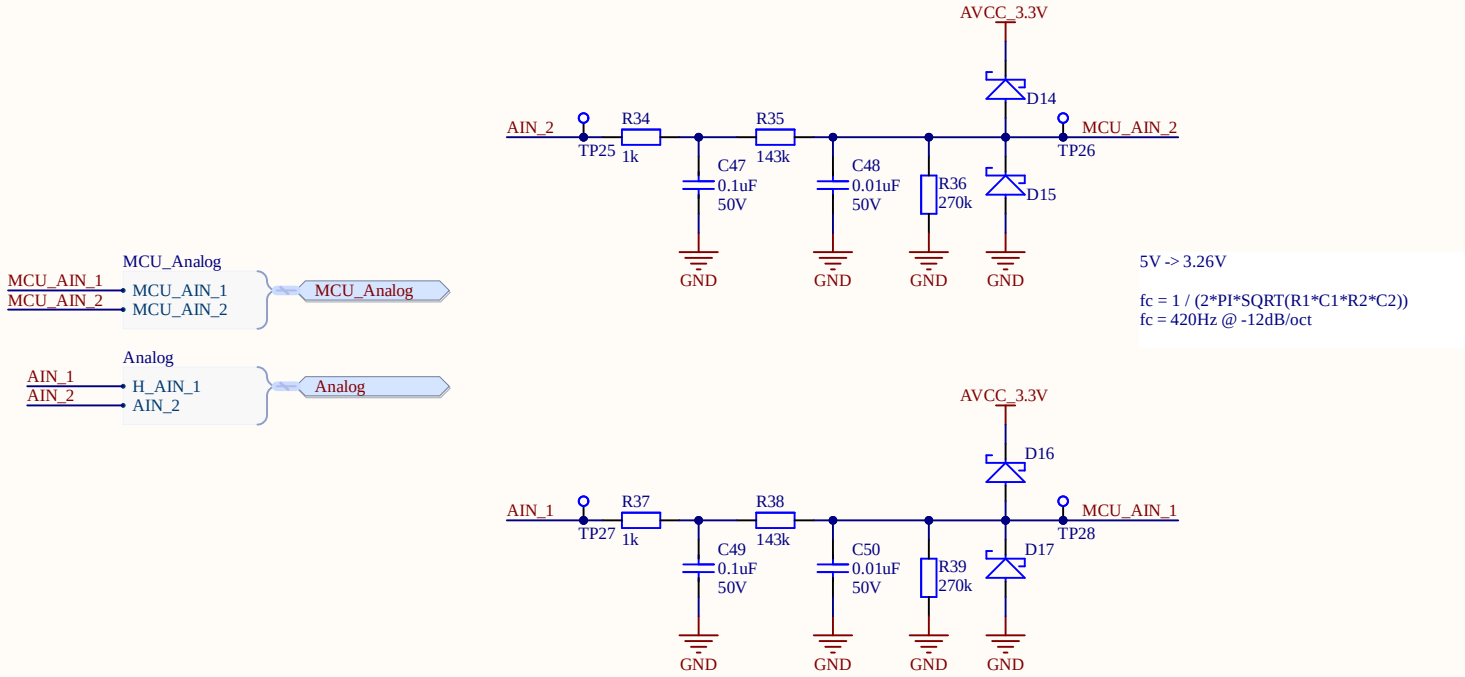
Male Female

Pin	Signal	Description
1	Reserved	No connection, only pass-through
2	CAN_L	CAN Low bus line (dominant low)
3	CAN_GND	CAN Ground (see grounding section)
4	Reserved	No connection, only pass-through
5	(CAN_SHLD)	No connection, only pass-through
6	(GND)	CAN Ground (see grounding section)
7	CAN_H	CAN High bus line (dominant high)
8	Reserved	No connection, only pass-through
9	(CAN_V+)	No connection, only pass-through

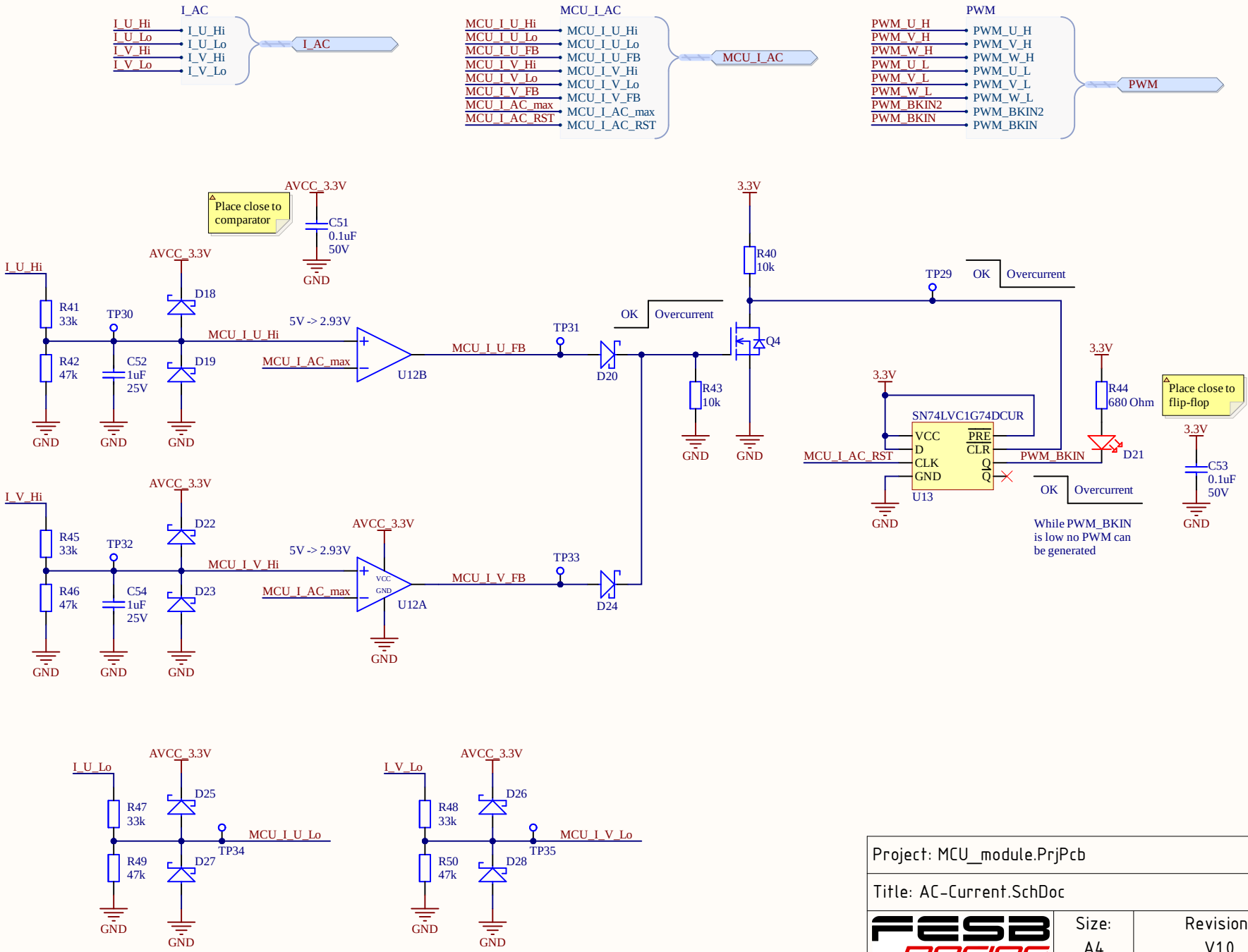
Project: MCU_module.PrjPcb		
Title: CAN.SchDoc		
FESB RACING	Size: A4	Revision: V1.0
Drawn by: Šimun Batovanja		Created: 16-Jul-25
Reviewed by: =CheckedBy		Edited: 06-Oct-25



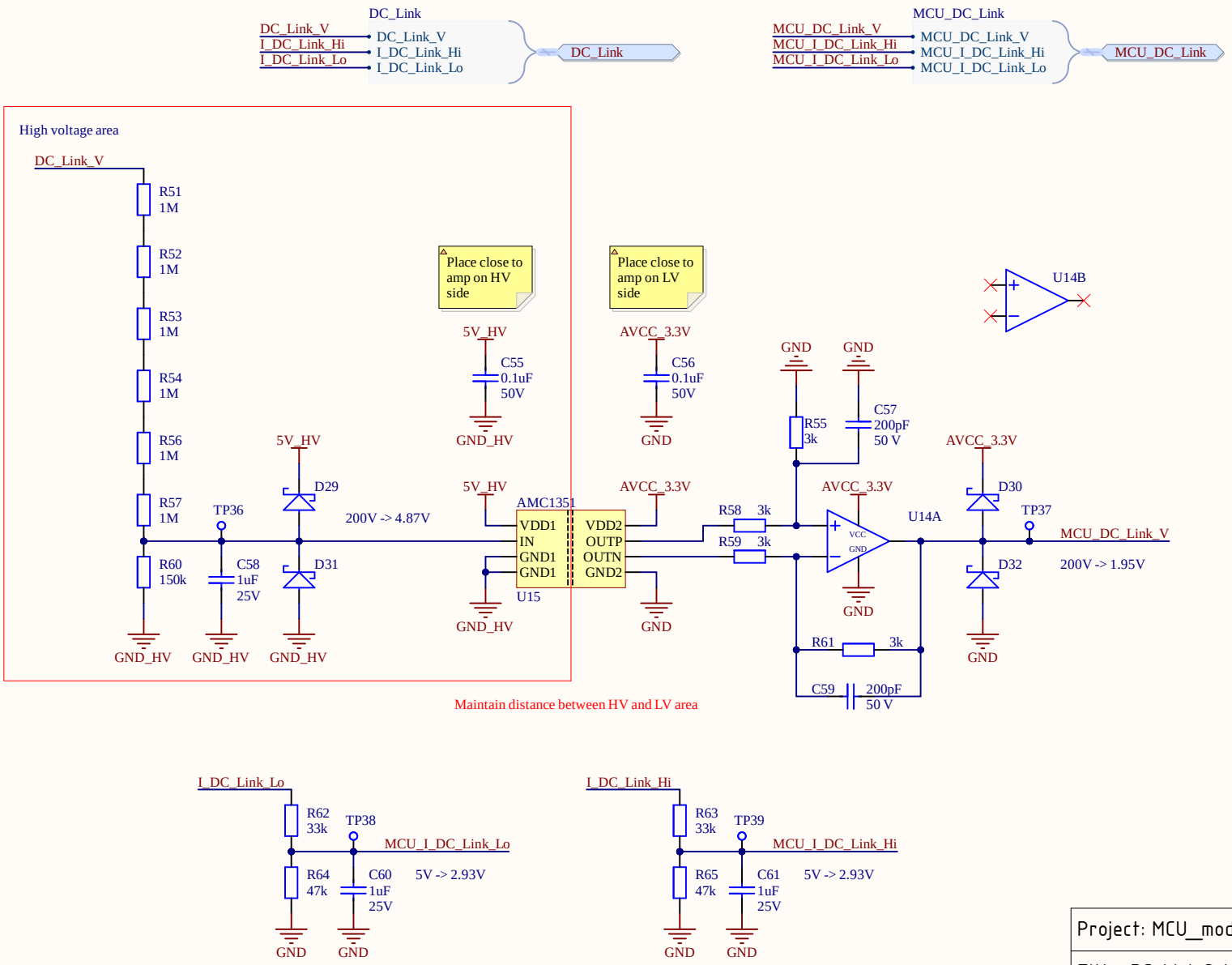
Project: MCU_module.PrjPcb		
Title: Digital-IO.SchDoc		
FESB RACING	Size: A4	Revision: V1.0
Drawn by: Šimun Batovanja		Created: 16-Jul-25
Reviewed by: =CheckedBy		Edited: 06-Oct-25



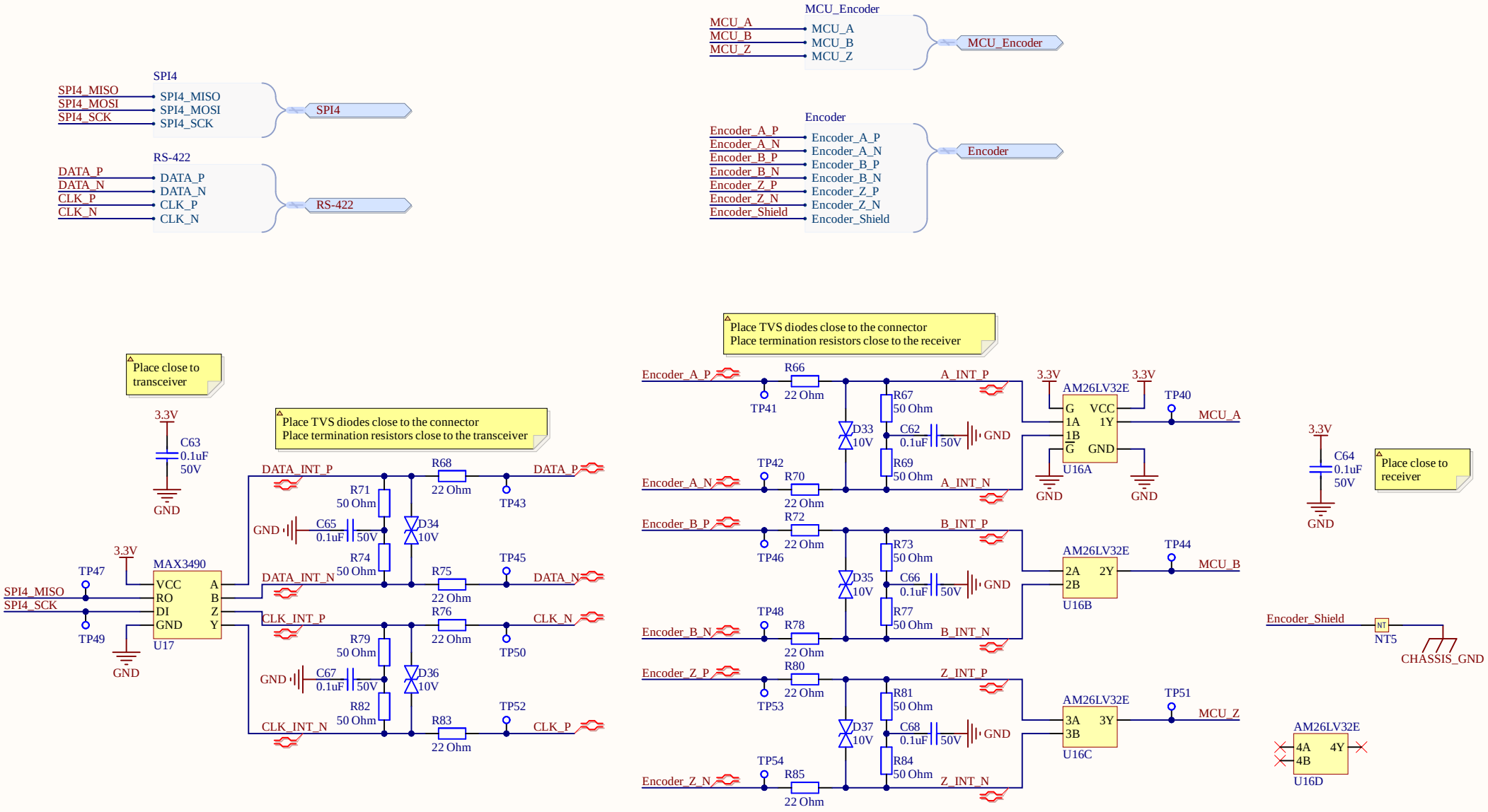
Project: MCU_module.PrjPcb		
Title: Analog-IN.SchDoc		
FESB RACING	Size: A4	Revision: V1.0
Drawn by: Šimun Batovanja		Created: 16-Jul-25
Reviewed by: =CheckedBy		Edited: 06-Oct-25



Project: MCU_module.PrjPcb		
Title: AC-Current.SchDoc		
FESB RACING	Size: A4	Revision: V1.0
Drawn by: Šimun Batovanja		Created: 16-Jul-25
Reviewed by: =CheckedBy		Edited: 06-Oct-25



Project: MCU_module.PrjPcb		
Title: DC-Link.SchDoc		
FESB RACING	Size: A4	Revision: V1.0
Drawn by: Šimun Batovanja		Created: 16-Jul-25
Reviewed by: =CheckedBy		Edited: 06-Oct-25



Project: MCU_module.PrjPcb	
Title: Encoder.SchDoc	
FESB RACING	Size: A4 / Revision: V1.0
Drawn by: Šimun Batovanja	Created: 16-Jul-25
Reviewed by: =CheckedBy	Edited: 06-Oct-25

Design requirement:
- 3.3V DAC output voltage
- 10Vrms output to resolver primary
- 10kHz output to resolver primary
- Bandwidth > 10 * f_out -> BW > 100kHz
- Limiting of DC excitation current

SR > 2 * PI * f * V_p
SR > 888.5 V/ms -> 888.5 mV/us

G = (2*V_p) / V_DACmax
G = 8.6

Resolver

Resolver_Sin_P
Resolver_Sin_N
Resolver_Cos_P
Resolver_Cos_N
Resolver_Exc_P
Resolver_Exc_N
Resolver_Shield

Resolver_Sin_P
Resolver_Sin_N
Resolver_Cos_P
Resolver_Cos_N
Resolver_Exc_P
Resolver_Exc_N
Resolver_Shield

Resolver

MCU_Resolver

MCU_Resolver_Sin
MCU_Resolver_Cos
MCU_Resolver_Exc

MCU_Resolver_Sin
MCU_Resolver_Cos
MCU_Resolver_Exc

MCU_Resolver

5V
C69
10uF
16V
GND

Place close to VCC

Resolver_Shield
NT
NT6
CHASSIS_GND

5V
TP56
MCU_Resolver_Exc
R88
10k
GND
U18A
VCC
GND
TP57
Resolver_Exc_P
GND

Resolver_Exc_N
NT
NT8
GND

U18B

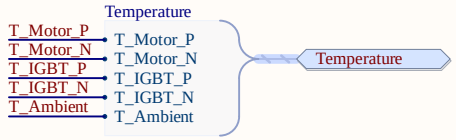
AVCC_3.3V
R86
680 Ohm
Resolver_Sin_N
R87
680 Ohm
GND

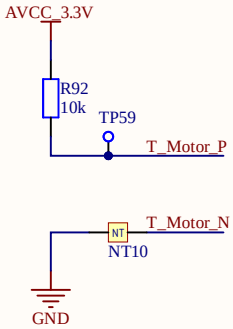
Resolver_Sin_P
NT
NT7
TP55
MCU_Resolver_Sin

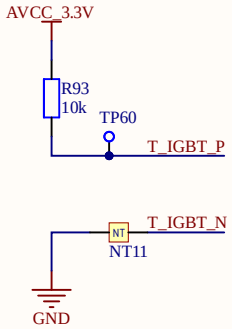
AVCC_3.3V
R89
680 Ohm
Resolver_Cos_N
R90
680 Ohm
GND

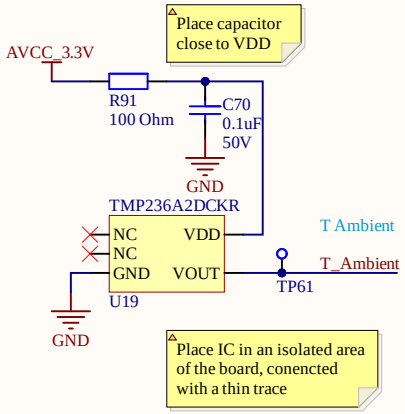
Resolver_Cos_P
NT
NT9
TP58
MCU_Resolver_Cos

Project: MCU_module.PrjPcb		
Title: Resolver.SchDoc		
FESB RACING	Size: A4	Revision: V1.0
Drawn by: Šimun Batovanja		Created: 16-Jul-25
Reviewed by: =CheckedBy		Edited: 06-Oct-25









Project: MCU_module.PrjPcb		
Title: Temperature-Meas.SchDoc		
FESB RACING	Size: A4	Revision: V1.0
Drawn by: Šimun Batovanja		Created: 25-Jul-25
Reviewed by: =CheckedBy		Edited: 06-Oct-25